Teacher Wu Kuan | See It at a Glance #27 — Absolute Value Inequalities on a Number Line

English lecture notes · YouTube Short _ubWLwWArZI

1. The trap question

Solve $|x - 3| < 2$.

2. The method

- $|x - a|$ is the distance from x to a on the number line.
- $|x - 3| < 2$ → within 2 of 3 → $1 < x < 5$ (in between).
- $|x - 3| > 2$ → more than 2 away → $x < 1$ or $x > 5$ (two outside parts).
- $|x + 3| = |x - (-3)|$ is the distance to -3. $\leq$ / $\geq$ include the endpoints.

3. Parent check method

Have your child mark the center and the radius on a number line before writing any inequality.

Practice

1. Solve $|x - 4| \leq 3$.
2. Solve $|x + 2| < 5$.
3. Solve $|2x - 1| \geq 5$.
4. How many integers x satisfy $|x - 1| < 4$?
5. Solve $|x - 1| + |x - 5| \leq 6$.

Answers

1. $1 \leq x \leq 7$
2. $-7 < x < 3$
3. $x \leq -2$ or $x \geq 3$
4. 7 ($x = -2, -1, 0, 1, 2, 3, 4$)
5. $0 \leq x \leq 6$

Saber Math · Teacher Wu Kuan (YouTube: Saber Math) · Free first-week trial of Kuan Weekly practice (email sign-up): https://kuan-weekly.vercel.app/sample · More: https://kuanmath.vercel.app · SAT notes: DM IG @saber_math